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127 Lung Abscess

Rebecca M. Baron, Beverly W. Baron, Miriam Baron Barshak

Lung abscess represents necrosis and cavitation of the lung following microbial infection. Lung abscesses can be single or multiple but usually are marked by a single dominant cavity >2 cm in diameter.

ETIOLOGY

The low prevalence of lung abscesses makes them difficult to study in randomized controlled trials. Although the incidence of lung abscesses has decreased in the antibiotic era, they are still a source of significant morbidity and mortality.

Lung abscesses usually are characterized as either primary (~80% of cases) or secondary. *Primary* lung abscesses generally arise from aspiration, often are caused principally by anaerobic bacteria, and occur in the absence of an underlying pulmonary or systemic condition. *Secondary* lung abscesses arise in the setting of an underlying condition, such as a postobstructive process (e.g., a bronchial foreign body or tumor) or a systemic process (e.g., HIV infection or another immunocompromising condition). Lung abscesses can also be characterized as acute (<4–6 weeks in duration) or chronic (~40% of cases).

EPIDEMIOLOGY

The majority of the existing epidemiologic information involves primary lung abscesses. In general, middle-aged men are more commonly affected than middle-aged women. The major risk factor for primary lung abscesses is aspiration. Patients at particular risk for aspiration, such as those with altered mental status, alcoholism, drug overdose, seizures, bulbar dysfunction, prior cerebrovascular or cardiovascular events, or neuromuscular disease, are most commonly affected. At additional risk are patients with esophageal dysmotility or esophageal lesions (strictures or tumors) and those with gastric distention and/or gastroesophageal reflux, especially those who spend substantial time in the recumbent position.

It is widely thought that colonization of the gingival crevices by anaerobic bacteria or microaerophilic streptococci (especially in patients with gingivitis and periodontal disease), combined with a risk of aspiration, is important in the development of lung abscesses. In fact, many physicians consider it extremely rare for lung abscesses to develop in the absence of teeth as a nidus for bacterial colonization.

The importance of these risk factors in the development of lung abscesses is highlighted by a significant reduction in abscess incidence in the late 1940s that coincided with a change in oral surgical technique: beginning at that time, these operations were no longer performed with the patient in the seated position without a cuffed endotracheal tube, and the frequency of perioperative aspiration events was thus decreased. In addition, the introduction of penicillin around the same time significantly reduced the incidence of and mortality rate from lung abscess.

PATHOGENESIS

Primary Lung Abscesses The development of primary lung abscesses is thought to originate when chiefly anaerobic bacteria (as well as microaerophilic streptococci) in the gingival crevices are aspirated into the lung parenchyma in a susceptible host **(Table 127-1)**. Patients who develop primary lung abscesses usually carry an overwhelming burden of aspirated material or are unable to clear the bacterial load. Pneumonitis develops initially (exacerbated in part by tissue damage caused by gastric acid); then, over a period of 7–14 days, the anaerobic bacteria produce parenchymal necrosis and cavitation whose extent depends on host–pathogen interaction **(Fig. 127-1)**. Anaerobes are thought to produce more extensive tissue necrosis in polymicrobial infections in which virulence factors of the various bacteria can act synergistically to cause more significant tissue destruction.

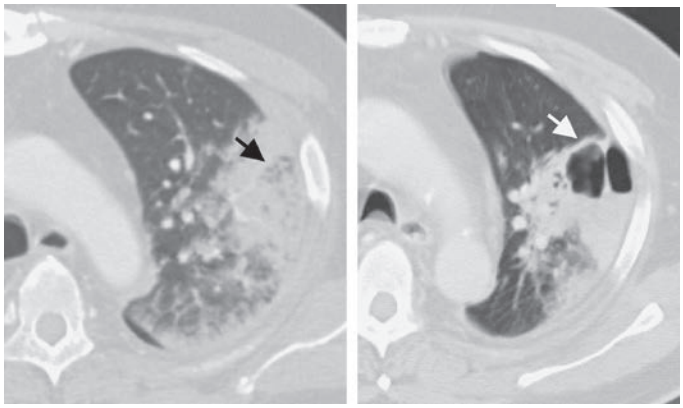
Secondary Lung Abscesses The pathogenesis of secondary abscesses depends on the predisposing factor. For example, in cases of bronchial obstruction from malignancy or a foreign body, the obstructing lesion prevents clearance of oropharyngeal secretions, leading to abscess development. With underlying systemic conditions (e.g., immunosuppression after bone marrow or solid organ transplantation), impaired host defense mechanisms lead to increased susceptibility to the development of lung abscesses caused by a broad range of pathogens, including opportunistic organisms (Table 127-1).

Lung abscesses also arise from septic emboli, either in tricuspid valve endocarditis (often involving *Staphylococcus aureus*) or in Lemierre's syndrome, in which an infection begins in the pharynx (classically involving *Fusobacterium necrophorum*) and then spreads

TABLE 127-1 Examples of Microbial Pathogens That can Cause Lung Abscesses

CLINICAL CONDITION	PATHOGENS
Primary lung abscess (usually with risk factors for aspiration)	Anaerobes (e.g., *Peptostreptococcus* spp., *Prevotella* spp., *Bacteroides* spp., *Streptococcus milleri*), microaerophilic streptococci
Secondary lung abscess (often with underlying immunocompromise)	*Staphylococcus aureus*, gram-negative rods (e.g., *Pseudomonas aeruginosa*, Enterobacteriaceae), *Nocardia* spp., *Aspergillus* spp., Mucorales, *Cryptococcus* spp., *Legionella* spp., *Rhodococcus equi*, *Pneumocystis jirovecii*
Embolic lesions	*Staphylococcus aureus* (often from endocarditis), *Fusobacterium necrophorum* (Lemierre's syndrome; see text for details)
Endemic infections (with or without underlying immunocompromise)	*Mycobacterium tuberculosis* (as well as *Mycobacterium avium* and *Mycobacterium kansasii*), *Coccidioides* spp., *Histoplasma capsulatum*, *Blastomyces* spp., parasites (e.g., *Entamoeba histolytica*, *Paragonimus westermani*, *Strongyloides stercoralis*)
Miscellaneous conditions	Bacterial pathogen (often *S. aureus*) after influenza or another viral infection, *Actinomyces* spp.

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A B

FIGURE 127-1 Representative chest CT scans demonstrating development of lung abscesses. This patient was immunocompromised by underlying lymphoma and developed severe *Pseudomonas aeruginosa* pneumonia, as represented by a left lung infiltrate with concern for central regions of necrosis (*panel **A**, black arrow*). Two weeks later, areas of cavitation with air-fluid levels were visible in this region and were consistent with the development of lung abscesses (*panel **B**, white arrow*). *(Images provided by Dr. Ritu Gill, Division of Chest Radiology, Brigham and Women's Hospital, Boston; with permission.)*

to the neck and the carotid sheath (which contains the jugular vein) to cause septic thrombophlebitis.

PATHOLOGY AND MICROBIOLOGY

Primary Lung Abscesses The dependent segments (posterior upper lobes and superior lower lobes) are the most common locations of primary lung abscesses, given the predisposition of aspirated materials to be deposited in these areas. Generally, the right lung is affected more commonly than the left because the right mainstem bronchus is less angulated.

Primary lung abscesses often are polymicrobial, primarily including anaerobic organisms as well as microaerophilic streptococci (Table 127-1). The retrieval and culture of anaerobes can be complicated by the contamination of samples with microbes from the oral cavity, the need for expeditious transport of the cultures to the laboratory, the need for early plating with special culture techniques, the prolonged time required for culture growth, and the need for collection of specimens prior to administration of antibiotics. When attention is paid to these factors, rates of recovery of specific isolates are reportedly as high as 78%.

Because it is not clear that knowing the identity of the causative anaerobic isolate alters the response to treatment of a primary lung abscess, practice has shifted away from the use of specialized techniques to obtain material for culture, such as transtracheal aspiration and bronchoalveolar lavage with protected brush specimens that allow recovery of culture material while avoiding contamination from the oral cavity. When no pathogen is isolated from a primary lung abscess (which occurs as often as 40% of the time), the abscess is termed a *nonspecific lung abscess*, and the presence of anaerobes is often presumed. A *putrid lung abscess* refers to cases with foul-smelling breath, sputum, or empyema; these manifestations are essentially diagnostic of an anaerobic lung abscess.

Secondary Lung Abscesses The location of secondary abscesses may vary with the underlying cause. The microbiology of secondary lung abscesses can encompass a broad bacterial spectrum, with infection by *Pseudomonas aeruginosa* and other gram-negative rods the most common. In addition, a broad array of pathogens can be identified in patients from certain endemic areas and in specific clinical scenarios (e.g., a significant incidence of fungal infections among immunosuppressed patients following bone marrow or solid organ transplantation). Because immunocompromised hosts and patients without the classic presentation of a primary lung abscess can be infected with a wide array of unusual organisms (Table 127-1), it is of special importance to obtain culture material in order to target therapy.

CLINICAL MANIFESTATIONS

Clinical manifestations initially may be similar to those of pneumonia, with fevers, cough, sputum production, and chest pain; a more chronic and indolent presentation that includes night sweats, fatigue, and anemia is often observed with anaerobic lung abscesses. A subset of patients with putrid lung abscesses may report discolored phlegm and foul-tasting or foul-smelling sputum. Patients with lung abscesses due to non-anaerobic organisms, such as *S. aureus*, may present with a more fulminant course characterized by high fevers and rapid progression.

Findings on physical examination may include fevers, poor dentition, and/or gingival disease as well as amphoric and/or cavernous breath sounds on lung auscultation. Additional findings may include digital clubbing and the absence of a gag reflex.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of lung abscesses is broad and includes other noninfectious processes that result in cavitary lung lesions, including lung infarction, malignancy, sequestration, cryptogenic organizing pneumonia, sarcoidosis, vasculitides and autoimmune diseases (e.g., granulomatosis with polyangiitis), lung cysts or bullae containing fluid, and septic emboli (e.g., from tricuspid valve endocarditis). Other less common entities can include pulmonary manifestations of diseases that usually present at locations other than the chest (e.g., inflammatory bowel disease, pyoderma gangrenosum).

DIAGNOSIS

Lung abscesses are documented by chest imaging. Although a chest radiograph usually detects a thick-walled cavity with an air-fluid level, CT permits better definition and may provide earlier evidence of cavitation. CT may also yield additional information regarding a possible underlying cause of lung abscess, such as malignancy, and may help distinguish a peripheral lung abscess from a pleural infection. This distinction has important implications for treatment because a pleural space infection, such as an empyema, may require urgent drainage.

As described earlier (see "Pathology and Microbiology," above), more invasive diagnostics (such as transtracheal aspiration) were traditionally undertaken for primary lung abscesses, whereas empirical therapy that includes drugs targeting anaerobic organisms currently is used more often. While sputum can be collected noninvasively for Gram's stain and culture, which may yield a pathogen, the infection is likely to be polymicrobial, and culture results may not reflect the presence of anaerobic organisms. Increasing use of molecular techniques for bacterial detection (e.g., 16S RNA gene amplification) may eventually yield increased specific pathogen identification. As stated above, many physicians consider putrid-smelling sputum to be virtually diagnostic of an anaerobic infection.

When a secondary lung abscess is present or empirical therapy fails to elicit a response, sputum and blood cultures are advised in addition to serologic studies for opportunistic pathogens (e.g., viruses and fungi causing infections in immunocompromised hosts). Additional diagnostics, such as bronchoscopy with bronchoalveolar lavage or protected brush specimen collection and CT-guided percutaneous needle aspiration, can be undertaken. Risks posed by these more invasive diagnostics include spillage of abscess contents into the other lung (with bronchoscopy) and pneumothorax and bronchopleural fistula development (with CT-guided needle aspiration). However, early diagnostics in secondary abscesses, especially in immunocompromised hosts, are particularly important because the patients involved may be especially fragile, at risk for infection with a broad array of pathogens, and therefore less likely than other patients to respond to empirical therapy.

TREATMENT

Lung Abscess

The availability of antibiotics in the 1940s and 1950s established therapy with this drug class as the primary approach to the treatment of lung abscess. Previously, surgery had been relied upon

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much more frequently. For many decades, penicillin was the antibiotic of choice for primary lung abscesses in light of its anaerobic coverage; however, because oral anaerobes can produce β-lactamases, clindamycin has proved superior to penicillin in clinical trials. For primary lung abscesses, the recommended regimens are (1) clindamycin (600 mg IV three times daily; then, with the disappearance of fever and clinical improvement, 300 mg PO four times daily) or (2) an IV-administered β-lactam/β-lactamase combination, followed—once the patient's condition is stable—by orally administered amoxicillin-clavulanate. This therapy should be continued until imaging demonstrates that the lung abscess has cleared or regressed to a small scar. Treatment duration may range from 3–4 weeks to as long as 14 weeks. One small study suggested that moxifloxacin (400 mg/d PO) is as effective and well tolerated as ampicillin-sulbactam. Notably, metronidazole is not effective as a single agent: it covers anaerobic organisms but not the microaerophilic streptococci that are often components of the mixed flora of primary lung abscesses.

In secondary lung abscesses, antibiotic coverage should be directed at the identified pathogen, and a prolonged course (until resolution of the abscess is documented) is often required. Treatment regimens and courses vary widely, depending on the immune state of the host and the identified pathogen. Other interventions may be necessary as well, such as relief of an obstructing lesion or treatment directed at the underlying condition predisposing the patient to lung abscess. Similarly, if the condition of patients with presumed primary lung abscess fails to improve, additional studies to rule out an underlying predisposing cause for a secondary lung abscess are indicated.

Although it can take as long as 7 days for patients receiving appropriate therapy to defervesce, as many as 10–20% of patients may not respond at all, with continued fevers and progression of the abscess cavity on imaging. An abscess >6–8 cm in diameter is less likely to respond to antibiotic therapy without additional interventions. Options for patients who do not respond to antibiotics and whose additional diagnostic studies fail to identify an additional pathogen that can be treated include surgical resection and percutaneous drainage of the abscess, especially when the patient is a poor surgical candidate. Timing of surgical intervention can be challenging; the goal is to balance the morbidity/mortality risk of a procedure with the need for definitively clearing the abscess in the setting of persistent infection that is not responsive to nonsurgical approaches. Possible complications of percutaneous drainage include bacterial contamination of the pleural space as well as pneumothorax and hemothorax.

COMPLICATIONS

Larger cavity size on presentation may correlate with the development of persistent cystic changes (pneumatoceles) or bronchiectasis. Additional possible complications include recurrence of abscesses despite appropriate therapy, extension to the pleural space with development of empyema, life-threatening hemoptysis, and massive aspiration of lung abscess contents.

PROGNOSIS AND PREVENTION

Reported mortality rates for primary abscesses have been as low as 2%, while rates for secondary abscesses are generally higher—as high as 75% in some case series. Other poor prognostic factors include age >60, the presence of aerobic bacteria, sepsis at presentation, symptom duration of >8 weeks, and abscess size >6 cm.

Mitigation of underlying risk factors may be the best approach to prevention of lung abscesses, with attention directed toward airway protection, oral hygiene, and minimized sedation with elevation of the head of the bed for patients at risk for aspiration. Prophylaxis against certain pathogens in at-risk patients (e.g., recipients of bone marrow or solid organ transplants or patients whose immune systems are significantly compromised by HIV infection) may be undertaken.

APPROACH TO THE PATIENT

Lung Abscess

For patients with a lung abscess and a low likelihood of malignancy (e.g., smokers <45 years old) and with risk factors for aspiration, it is reasonable to administer empirical treatment and then to pursue further evaluation if therapy does not elicit a response. However, some clinicians may opt for up-front cultures, even in primary lung abscesses. In patients with risk factors for malignancy or other underlying conditions (especially immunocompromised hosts) or with an atypical presentation, earlier diagnostics should be considered, such as bronchoscopy with biopsy or CT-guided needle aspiration. Bronchoscopy should be performed early in patients whose history, symptoms, or imaging findings are consistent with possible bronchial obstruction. In patients from areas endemic for tuberculosis or patients with other risk factors for tuberculosis (e.g., underlying HIV infection), induced sputum samples should be examined early in the workup to rule out this disease.

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128 Infective Endocarditis

Sara E. Cosgrove, Adolf W. Karchmer

The prototypic lesion of infective endocarditis (IE), the *vegetation* (Fig. 128-1), is a mass of platelets, fibrin, microorganisms, and scant inflammatory cells. Infection most commonly involves heart valves but may also occur on the low-pressure side of a ventricular septal defect, on mural endocardium damaged by aberrant jets of blood or foreign bodies, or on intracardiac devices. The analogous process involving arteriovenous shunts, arterio-arterial shunts (patent ductus arteriosus), or a coarctation of the aorta is called *infective endarteritis*.

IE can be classified according to the temporal evolution of disease, the site of infection, the cause of infection, or the predisposing risk factor (e.g., injection drug use, health care–associated). *Acute IE* is a hectically febrile illness that rapidly damages cardiac structures, seeds extracardiac sites, and, if untreated, progresses to death within weeks. *Subacute IE* follows an indolent course; causes structural cardiac damage only slowly, if at all; rarely metastasizes; and is gradually progressive unless complicated by a major embolic event or a ruptured mycotic aneurysm.

In the United States and likely in other developed countries, the incidence of IE is estimated to be 12 cases per 100,000 population per